

Refinery Process Units

CDU

Crude Distillation Unit

Separate crude oil into lighter and heavier fractions

RHU

Residu Hydrocracking Unit

Converts vacuum residue into lighter products with hydrogen

VDU

Vacuum Distillation Unit

Vacuum distillation to separate heavier fractions

SRU

Sulfur Recovery Unit

Recovers sulfur from acid gases (H_2S)

FCC

Fluid Catalytic Cracking

Fluid catalytic cracking produces lighter fuels and propylene

RHU

Residue Hydrocracking Unit

Converts sulfur into lighter products with hydrogen

NHT

Naphtha Hydrotreating

Removes sulfur and nitrogen compounds from naphtha

ISOM

Isomerization Unit

Boosts the octane of light naphtha

PRU

Propane Recovery Unit

Separates and recovers propane from gases

ARU

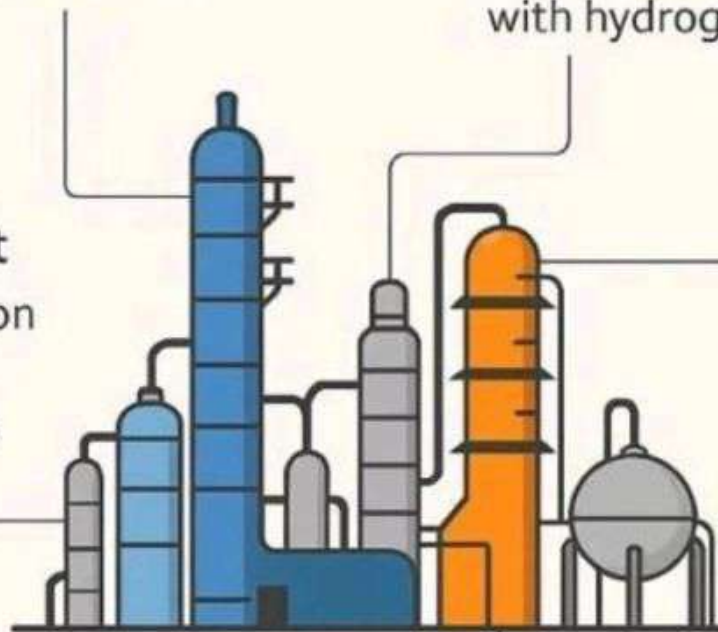
Amine Recovery Unit

Absorbs acid gases with amine for recovery

DHDS

Diesel Hydrodesulfurization

Reduces sulfur in diesel for compliance





KEY PROCESS UNITS OF AN OIL REFINERY

Transforming Crude Oil into Energy. Powering the World.



CDU

Crude Distillation Unit

Initial Separation
of Crude Oil



Separates crude into different
fractions based on boiling range.

VDU

Vacuum Distillation Unit

Refining
Heavy Residues



Processes heavy residues under
vacuum to prevent cracking.

FCC

Fluid Catalytic
Cracking

Converts
Heavy Oils
to Gasoline



Breaks heavy molecules into lighter,
high-value products like gasoline.

Hydrotreating

NHT / DHT

Removes Sulfur
& Impurities



Purifies products by removing sulfur,
nitrogen, and other impurities.

H₂

CCR

Catalytic Reforming

Boosts Octane
in Fuels



Transforms low-octane naphtha
into high-octane gasoline.

Alkylation Unit

Produces
High-Octane
Gasoline



Combines olefins with isobutane
to produce premium alkylate.

SRU

Sulfur Recovery Unit

Recovers
Elemental
Sulfur



Recovers sulfur from gases
to reduce environmental impact.

ARU

Amine Recovery Unit

Removes H₂S
from Gases



Eliminates hydrogen sulfide
for safe and clean operations.

H₂S

ARU

Amine Recovery Unit

Removes H₂S
from Gases



Regenerates amine solutions
by removing absorbed impurities.

H₂S

Delayed Coking

Converts Residue
to Coke & Products



Converts heavy residue into valuable
products and petroleum coke.

Isomerization Unit

Increases Octane
in Light Naphtha



Improves fuel quality by converting
n-paraffins to isoparaffins.

